

Task 12: Support Vector Machines (SVM) - Interview Questions & Answers

Q: What is a Hyperplane?

A: A hyperplane is a decision boundary used by SVM to separate different classes in the feature space.

Q: What is the Kernel Trick?

A: The kernel trick transforms data into a higher-dimensional space so that non-linear data can be separated using a linear classifier.

Q: What is the difference between Hard Margin and Soft Margin?

A: Hard Margin allows no misclassification and works only for perfectly separable data. Soft Margin allows some misclassification to improve generalization.

Q: What is the role of C in SVM?

A: C is the regularization parameter. A high C tries to classify all training examples correctly, while a low C allows more flexibility.

Q: What is Gamma in SVM?

A: Gamma determines the influence range of a single training sample. High gamma may lead to overfitting, while low gamma gives smoother decision boundaries.

Q: Why is Feature Scaling important in SVM?

A: SVM relies on distance calculations, so scaling ensures that all features contribute equally.

Q: What are the common kernels used in SVM?

A: Linear, Polynomial, Radial Basis Function (RBF), and Sigmoid kernels.

Q: When should you use a Linear Kernel?

A: When the data is approximately linearly separable and has many features.

Q: What is the RBF Kernel?

A: The RBF kernel maps data into a higher-dimensional space and is commonly used for non-linear classification.

Q: How do you evaluate an SVM model?

A: Using metrics such as Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC Curve.